

Lecture 5: Hierarchical Modeling

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Announcements

- Reading: Chapter 5 of BDA

- HW 2 due Sunday.

$$Y = CZ$$

$$Z \sim N(0, I)$$

Scale mixtures

$$C \sim \text{Exp}(1/2) \Rightarrow Y \sim \text{Laplace}$$

$$C \sim \text{Inv-Gam} \Rightarrow Y \sim \text{t-distr.}$$

scalar

$$Z \sim N_p(0, I) \quad Z \perp X$$

$$Y_p \sim \sqrt{\frac{2}{X}} Z_p$$

Gamma

$\Rightarrow$ Multivariate
t-distribution

Comparing Multiple Related Groups

- Hierarchy of nested populations
- Models which account for this are called *hiearchical* or *multi-level* models

Some examples:

- Patient outcomes within several different hospitals
- People within counties in the United States (e.g. Asthma mortality example)
- Athlete performance in sports
- Genes within a group of animals

Eight schools example

- A study was performed for the Educational Testing Service (ETS) to evaluate the effects of coaching programs on SAT preparation
- Each of eight different schools used a short-term SAT prep coaching program
- Compute the average SAT score in those who did take the program minus those that did not participate in the program
- We observe the average difference varies by school. What accounts for these differences?

Observe $y_1, \dots, y_8, \sigma_1, \dots, \sigma_8$ related to sample size.

Why are $Y_1, Y_2, \dots, Y_8$ different numbers?

- Wealth of students in school.
- Absenteeism, ESL. Demographics
- Teacher Resources.
- Sampling Variability.

$$Y_j \sim N(\theta_j, \sigma_j^2) \quad j=1, \dots, 8$$

Want θ 's, Observe Y 's.

Y 's mix "signal" & "noise"

Eight schools example

- Interested in “real” differences due to training
- Want to reduce ^{impact} ~~effect~~ ^{“sampling”} of chance variability
- How do we estimate the effect of the program in each of the schools?

θ_i ?

θ_i

Eight schools example

- Consider two extremes:

1.
 - Estimate the effect of the program in every school independently *8 different analyses*
 - A separate prior distribution for each school effect
 - Or assume the effect is the same in every school
 - Combine all the data *$\theta_1 = \theta_2 = \dots = \theta_8$*
3.
 - A compromise between the above 2 options?

1. $Y_j \sim N(\theta_j, \sigma_j^2)$

$p(\theta_j)$ (separate prior)

2. $Y_j \sim N(\theta, \sigma_j^2)$ (shared θ)

Eight Schools Example

$$y_j \sim N(\theta_j, \sigma_j^2)$$

- y_j is the observed effects of the program in school j
 - Based on a sample of test scores from those in the program and those not in the program
- θ_j are the true *unknown* effects of the program in school j
- Assume variances, σ_j^2 , are *known*
 - e.g. determined by the number of students in the sample

Eight Schools Example

```
1 J <- 8  
2 y = c(28, 8, -3, 7, -1, 1, 18, 12)  
3 sigma <- c(15, 10, 16, 11, 9, 11, 10, 18)
```

- Assuming the effect of the program on each school is identical.
- What are the chances of seeing a value as large as 28?
- As small as -3?

Eight Schools Example

- Assume the effect of the program on each school is identical,

i.e. $\theta_j = \mu$

$p(\mu) \propto \text{const}$ (improper)

- Assume a flat prior on μ , what is

$p(\mu \mid y_1, \dots, y_8, \sigma_1, \dots, \sigma_8)$?

$P(\mu \mid y_1, \dots, y_8, \sigma_1, \dots, \sigma_8) \propto L(\mu) \propto$

$\prod_{i=1}^8 \frac{1}{\sqrt{2\pi}\sigma_i}$

$e^{-\frac{(y_i - \mu)^2}{2\sigma_i^2}}$

$\propto e^{-\sum \frac{1}{2\sigma_i^2} (y_i - \mu)^2}$

$$\propto e^{-\sum \frac{1}{2\sigma_i^2} (\mu^2 - 2y_i\mu)}$$

$$\propto e^{-\sum \frac{1}{2\sigma_i^2} \left(\mu - \frac{\sum \frac{y_i}{\sigma_i^2}}{\sum 1/\sigma_i^2} \right)^2}$$

Check

$$\Rightarrow N\left(\sum w_i y_i, \frac{1}{\sum 1/\sigma_i^2}\right)$$

$$w_i = \frac{1/\sigma_i^2}{\sum 1/\sigma_i^2}$$

"Fisher Weighting"
Fisher Info for μ .

$$\hat{\mu}_{MLE} = \sum w_i y_i$$

Eight Schools Example

```
1  ## Compute the precision from each school
2  prec <- 1/sigma^2
3
4  ## global estimate is a weighted vareage
5  mu_global <- sum(prec * y / sum(prec))
6  mu_global
```

```
[1] 7.685617
```

||
 μ

Eight Schools Example

- Assume the effect of the program on each school is identical, i.e. $\theta_j = \mu = 7.7$?
- What are the chances of school 1 having an effect large as 28 (given $\sigma_1 = 15$)?
- Y_3 as small as -3 (given $\sigma_3 = 16$)?

$$P(\hat{Y} > 28 \mid y_1, \dots, y_8, \sigma_1, \dots, \hat{\sigma} = 15)$$

$$P(\hat{Y} < -3 \mid \dots, \hat{\sigma} = 16)$$

$$P(\tilde{Y} / y_1, \dots, y_8) = \int_{-\infty}^{\infty} P(\tilde{Y} / \mu) P(\mu / y_1, \dots, y_8) d\mu$$

$$P(\tilde{Y} / \mu) \sim N(\mu, \tilde{\sigma}^2)$$

$$P(\mu / y_1, \dots, y_8) \sim N(\sum w_i y_i, \frac{1}{\sum w_i \sigma_i^2})$$

$$\tilde{Y} = \underbrace{\mu}_{\substack{\downarrow \\ N(0, \tilde{\sigma}^2)}} + \underbrace{\varepsilon}_{\substack{\downarrow \\ N(0, \tilde{\sigma}^2)}} \Rightarrow N(\underbrace{\sum w_i y_i}_{\substack{\text{sample} \\ \text{mean}}}, \underbrace{\tilde{\sigma}^2 + \frac{1}{\sum w_i \sigma_i^2}}_{\substack{\text{uncertainty} \\ \text{in } \mu}})$$

$$\sigma_i^2 = \frac{\sigma^2}{n_i}$$

Posterior Prediction Under Complete Pooling

```
1 prec <- 1/sigma^2
2
3 ## global estimate is a weighted average
4 mu_global <- sum(prec * y / sum(prec))
5
6 print(sprintf("mu is %f", mu_global))
```

```
[1] "mu is 7.685617"
```

```
1 1 - pnorm(28, mean=mu_global, sd=sqrt(1/sum(1/sigma^2)+sigma[1]^2))
```

```
[1] 0.09560784
```

7.7

```
1 pnorm(-3, mean=mu_global, sd=sqrt(1/sum(1/sigma^2)+sigma[3]^2))
```

```
[1] 0.2587447
```


Eight Schools Example

$$y_j \sim N(\theta_j, \sigma_j^2)$$

- θ_j are the true unknown effects of the program in school j
- y_j is the observed effects of the program in school j
 - Based on a sample of test scores from those in the program and those not in the program
 - Number of people in the sample determine the magnitude of σ_j^2

Eight Schools Example

How do we estimate θ_j ?

- Assume effects are totally independent: $\hat{\theta}_j^{(MLE)} = y_j$ is the MLE

"No Pooling"

- Assume effects are identical: $\hat{\theta}_j^{(pool)} = \frac{\sum_i \frac{1}{\sigma_i^2} y_i}{\sum_i \frac{1}{\sigma_i^2}} = \mu$

- Same effect for all schools: estimate using a weighted average of the observed effects

Eight Schools

```
1 theta_j_mle <- y
2 theta_j_mle
```

```
[1] 28  8 -3  7 -1  1 18 12
```

```
1 theta_j_pooled <- rep(sum(1/sigma^2 * y) / sum(1/sigma^2), J)
2 theta_j_pooled
```

```
[1] 7.685617 7.685617 7.685617 7.685617 7.685617 7.685617 7.685617 7.685617
```

- Compromise: $\hat{\theta}_j^{\text{shrink}} = w \theta_j^{\text{MLE}} + (1 - w) \theta^{\text{pooled}}$

"Partial Pooling"

Eight schools example

Add a shared normal prior distribution to θ_j

$$\theta_i \sim N(\mu, \tau^2)$$

↓ assume known for now.

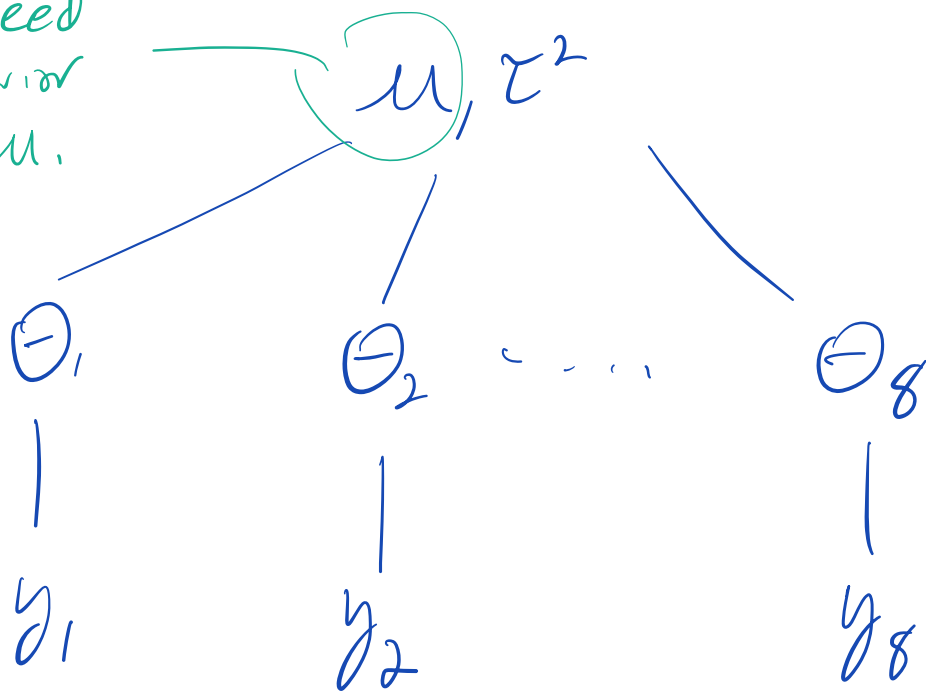
$$y_j \sim N(\theta_j, \sigma_j^2)$$

Need a prior on μ , "global mean"

- The global mean, μ , is also an unknown parameter. What prior should we choose?
- τ^2 determines how much weight we put on the independent estimate vs the pooled estimate.
- A 9-parameter posterior:

$$p(\mu, \theta_1, \dots, \theta_8 \mid y_1, \dots, y_8, \sigma_1, \dots, \sigma_8, \tau^2)$$

need
a prior
on μ .



If μ and τ^2 were known,
what is $P(\Theta_1 | y_1, \underline{y_2 \dots y_8}, \mu, \tau^2)$?

$$\sim \mathcal{N}(w y_1 + (1-w)\mu, \frac{1}{1/\sigma_1^2 + \frac{1}{\tau^2}})$$

$$w = \frac{1/\sigma_1^2}{1/\sigma_1^2 + 1/\tau^2} \quad \left(\begin{array}{l} \text{normal -} \\ \text{normal} \\ \text{model} \end{array} \right)$$

μ is not known

So, $y_2, \dots, y_n$ provide
info about θ_1 via
 μ .

Intuition:

$$\text{Var}(Y) = \text{Var}(\theta + \epsilon)$$

$\uparrow$ $\uparrow$
 $N(\mu, \tau^2)$ $N(0, \sigma^2)$

$$\tau^2 + \sigma^2$$

$$\text{Var}(\hat{\theta}) \approx \tau^2$$

Intuition for shrinkage

- $Y_j = \theta_j + \epsilon_j$
 - For simplicity assume $Var(\epsilon_j) = \sigma^2$ for all j
 - θ_j represents true effect in school j (signal)
 - $Var(\theta_j) = \tau^2$ represents how much the true effects vary across schools
 - ϵ_j is sampling variability (noise, chance variation)
- $\hat{\theta}_{MLE} = Y_j$

Intuition for shrinkage

- Consequence: the observed outcomes always have higher variance than the signal, i.e. $\text{Var}(Y_j) > \text{Var}(\theta_j)$
- Intuition: reduce the variance by shrinking estimates to a common mean!
- The variance of the shrunk estimates should be close to τ^2

μ (also unknown, though)

Eight schools example

Questions:

- Is the training program effective in school j ?
 - What is $P(\theta_j > 0 \mid y)$?
- On average (over all schools) is the training program effective?
 - What is $P(\mu > 0 \mid y)$?
- Will the training program be effective in a new school?
 - What is $P(\theta_{J+1} > 0 \mid y)$?

$$P(\tilde{\gamma}_{J+1} > 0 \mid \dots)$$

(predictive)

Eight schools example

Comments:

- The global average, μ , is a parameter so also has uncertainty
- How do we determine how much to shrink, e.g. how do we determine τ^2 ?
- What σ_j^2 were also unknown?

Eight schools example

$$\Theta \sim \mathcal{N}(\mu, \tau^2)$$

- If τ^2 is large, the prior for θ_j is not very strong
 - If $\tau^2 \rightarrow \infty$ equivalent to the no pooling model
- If τ^2 is small, we assume a priori that θ_j are very close
 - if $\tau^2 \rightarrow 0$ equivalent to the complete pooling model,
 $\theta_j = \mu$

$$\mathbb{E}[\Theta_i | \mu] = w_i y_i + (1 - w_i) \mu$$

$$w_i = \frac{1/\sigma_i^2}{1/\sigma_i^2 + 1/\tau^2}$$

Inference (Monte Carlo)

$$P(\theta_1, \dots, \theta_8, \mu | \dots)$$

- Factorize the density into tractable components

$$P(\mu | \dots) P(\theta | \mu, \dots)$$

1. Draw $\square p(\mu | y_1, \dots, y_8, \tau^2) \sim \mathcal{N}(\sum w_i y_i, \frac{1}{\sum 1/\sigma_i^2})$

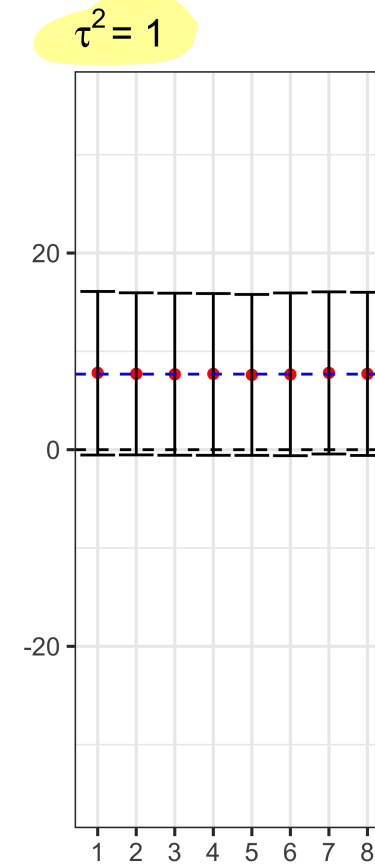
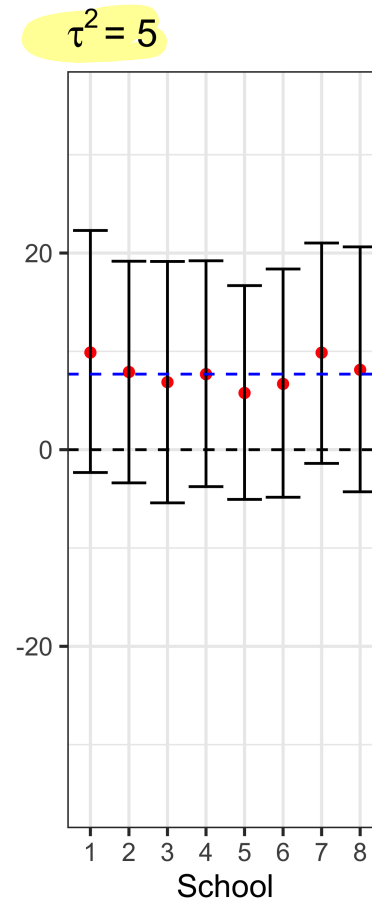
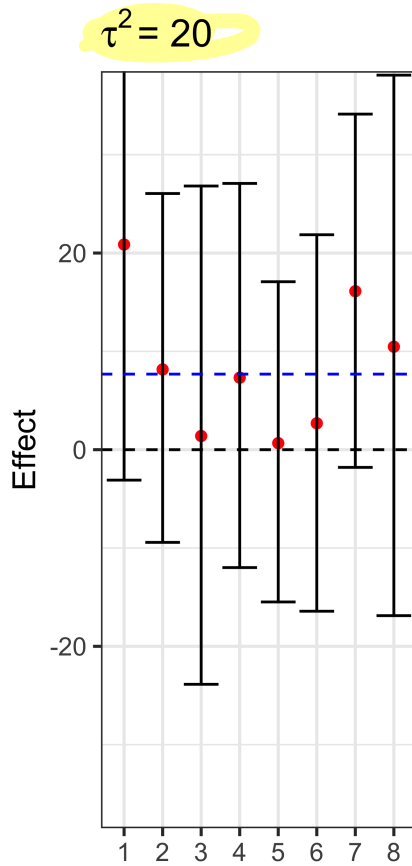
2. Draw $\square p(\theta_i | \mu, y_i, \tau^2) \sim \mathcal{N}(\frac{\frac{1/\sigma_i^2}{1/\sigma_i^2 + 1/\tau^2} y_i + \frac{1/\tau^2}{1/\sigma_i^2 + 1/\tau^2} \mu, \cdot)$

- Later: MCMC or other approximate methods

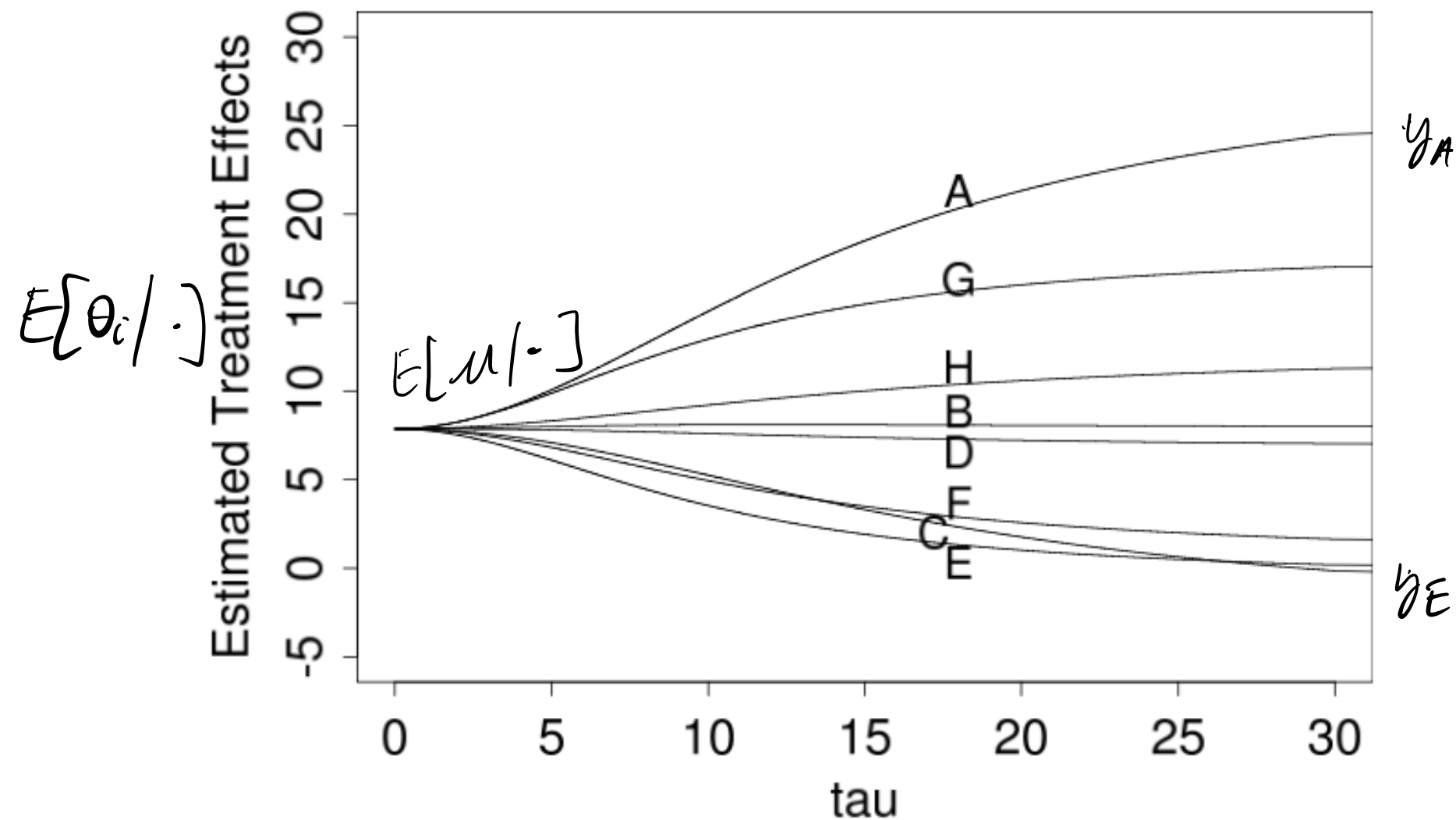


Eight Schools example

Notice
how
interval
lengths
differ.



The impact of τ



The impact of τ

